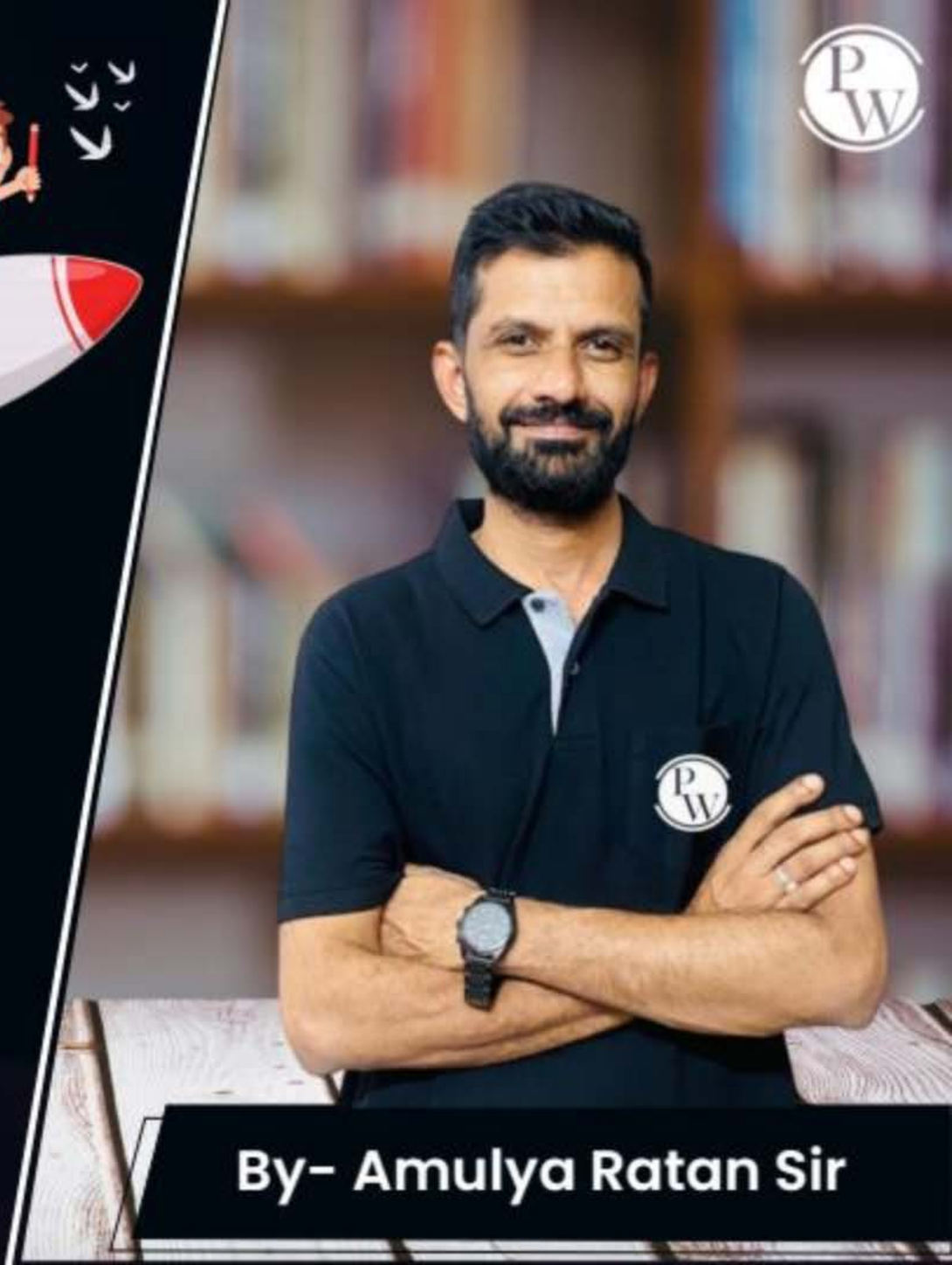
GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE



Lecture No.- 07

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Recap of Previous Lecture

1-1

Topic Number System

Prime

Non Prime OR Composite

Topics to be Covered

Topic-1 More on Number System

Topic-2 Time and Work

[MCQ]

#Q. How many zeroes would be at the end in the answer of 50! ?

$25 = 5 \times 5$

$25 \times 2 = 50$

Assignment

$50 \times 49 \times 48 \times 47 \times \ldots \ldots 2 \times 1$

$1 \times 2 \times 3 \times 4 \times \boxed{5} \times 6 \times 7 \times 8 \ldots \ldots 50$

1 – 10 → 2^2

11 – 20 → 2^2

21 – 30 → 3^2

31 – 40 → 2^2

41 – 50 → 3^2

12

A. 12

B. 15

C. 23

D. 24

$190! \rightarrow \boxed{4}$ 92

$2 \times 2 = 4$

$(2)^{184}$

$(4)^{92}$

$\frac{190}{2} \Rightarrow 95$

$\frac{95}{2} \Rightarrow 47$

$\frac{47}{2} \Rightarrow 23$

$\frac{23}{2} \rightarrow 11$

$\frac{11}{2} \rightarrow 5$

$\frac{5}{2} \rightarrow 2$

$\frac{2}{2} \rightarrow 1$

$\frac{1}{2} \rightarrow 0$

184 92

$\frac{50}{5} \rightarrow 10$

$\frac{10}{5} \rightarrow 2$

$\frac{2}{5} \Rightarrow 0$

12

$100! \rightarrow 6^{48}$

$2 \times 3 = 6$

$2 \rightarrow 97$

$3 \rightarrow 48$

$\frac{100}{2} = 50$

$\frac{50}{2} \rightarrow 25$

$\frac{25}{2} \rightarrow 12$

$\frac{12}{2} \rightarrow 6$

$\frac{6}{2} \rightarrow 3$

$\frac{3}{2} \rightarrow 1$

$\frac{1}{2} \rightarrow 0$

97

$\frac{100}{3} \rightarrow 33$

$\frac{33}{3} \rightarrow 11$

$\frac{11}{3} \rightarrow 3$

$\frac{3}{3} \rightarrow 1$

$\frac{1}{3} \rightarrow 0$

48

$$60 \rightarrow \boxed{12}^{2^8}$$

$$\boxed{12} \Rightarrow \boxed{2 \times 2} \times 3$$

$$\downarrow \qquad \downarrow$$

$$28 \qquad 28$$

$$\frac{60}{2} \rightarrow 30$$

$$\frac{1}{2} \rightarrow 0 \qquad \frac{30}{2} \rightarrow 15$$

$$\frac{15}{2} \rightarrow 7$$

$$\frac{7}{2} \rightarrow 3$$

$$\frac{3}{2} \rightarrow 1$$

$$58$$

$$\frac{60}{3} \rightarrow 20$$

$$\frac{20}{3} \Rightarrow 6$$

$$\frac{6}{3} \rightarrow 2$$

$$\frac{2}{3} \rightarrow \cancel{0}\ \cancel{28}$$

[MCQ]

#Q. If 50! can be denoted maximum 7^x, then the value of x is?

A. 9

B. 8

C. 7

D. None of these

$\frac{50}{7} \Rightarrow 7$

$\frac{7}{7} \Rightarrow 1$

$\frac{1}{7} \Rightarrow 0$

8

$50! = 7^8$

$1-10 \Rightarrow 1^0$

$11-20 \Rightarrow 1^0$

$21-30 \Rightarrow 2^0$

$31-40 \Rightarrow 1^0$

$41-50 \Rightarrow 3^0$

8

[MCQ]

#Q. Find the number of zeroes in following multiplication:

5 x 10 x 15 x 20 x 25..... x 50 ?

A 48

B 12

C 24

D 8

$\boxed{5 \times 2}$

12 8

$\frac{10}{5} \to 2$

$\frac{2}{5} \to 0$

$= 5 \times 1 \times 5 \times 2 \times 5 \times 3 \times 5 \times 4 \ldots\ldots 5 \times 10$

$= 5^{10} \times 10!$

$\frac{10}{2} \to 5$

$\frac{5}{2} \to 2$

$\frac{1}{2} \to 0$ $\frac{2}{2} \to 1$

8

P.4

$! \Rightarrow 9$

$\boxed{3 \times 3} = 9$

$$\begin{array}{r} 14 \\ 2\overline{)29} \\ 28 \\ \hline 1 \end{array}$$

$3 \rightarrow 28$

[MCQ]

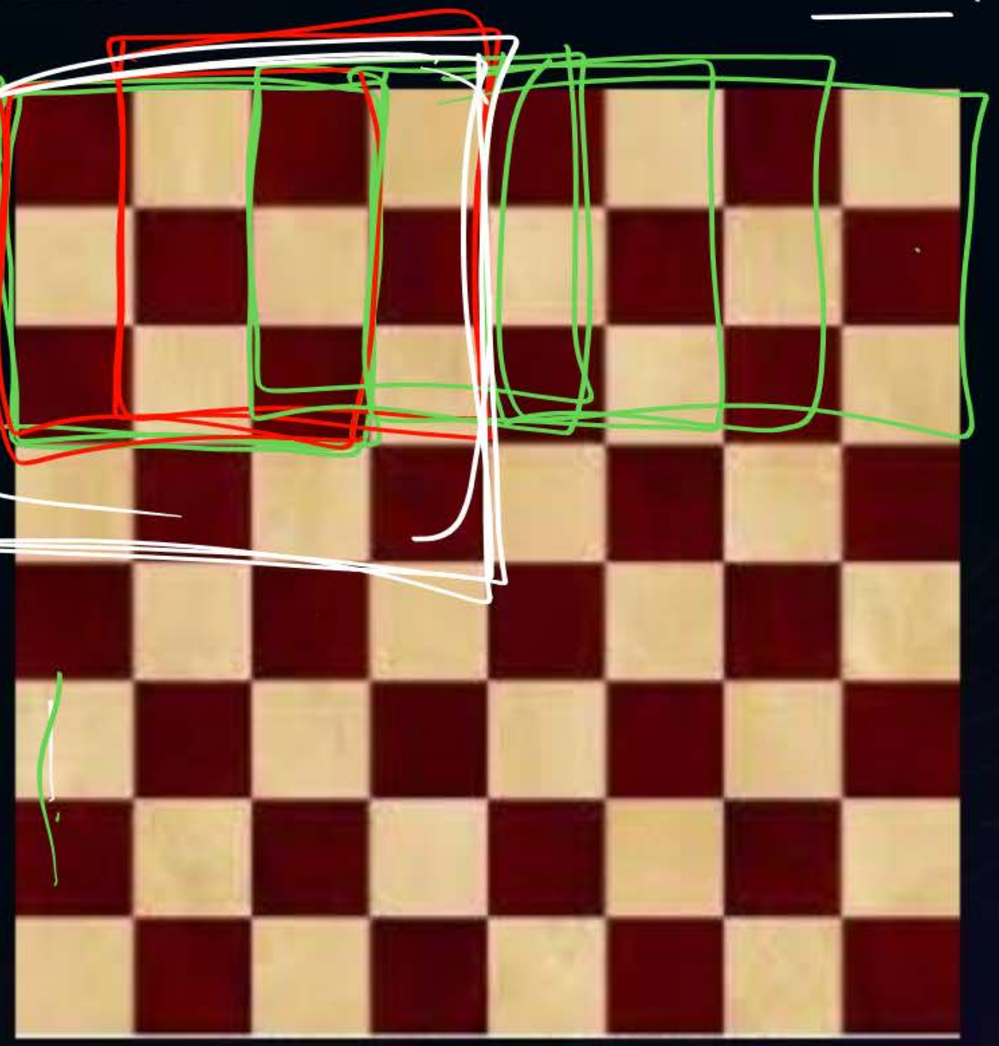
#Q. How many squares are there in a chessboard?

1x1 →	4x4 = 16
2x2 →	3x3 = 9
3x3 →	2x2 = 4
4x4 →	1x1 = 1

30 204

Size	No. 204
1x1 →	8x8 = 64
2x2 →	7x7 = 49
3x3 →	6x6 = 36
4x4 →	5x5 = 25
5x5 →	4x4 = 16
6x6 →	3x3 = 9
7x7 →	2x2 = 4
8x8 →	1x1 = 1

8 m



[MCQ]

#Q. What would be the unit digit in the answer of: $2362^{911} + 1418^{216}$

$4 \overline{)911}$ $\quad$ $4 \overline{)216}$

$8 + 6 \rightarrow 4$

64×8

$2^1 \rightarrow$	2	R=1
$2^2 \rightarrow$	4	R=2
$2^3 \rightarrow$	8	R=3
$2^4 \rightarrow$	6	R=0
$2^5 \rightarrow$	2	
$2^6 \rightarrow$	4	

$8^1 \rightarrow$	8	R=1
$8^2 \rightarrow$	4	R=2
$8^3 \rightarrow$	2	R=3
$8^4 \rightarrow$	6	R=0
$8^5 \rightarrow$	8	

[MCQ]

#Q. What would be the unit digit in the answer of: $2913^{1902} + 1647^{460}$?

$9 + 1 = 0$

$3^1 \rightarrow 3$

$3^2 \rightarrow 9$ R=2

$3^3 \rightarrow 7$

$3^4 \rightarrow 1$

$3^5 \rightarrow 3$

$7^1 \rightarrow 7$

$7^2 \rightarrow 9$

$7^3 \rightarrow 3$

$7^4 \rightarrow 1$ R=0

$7^5 \rightarrow 7$

A. 0

B. 2

C. 3

D. 5

[MCQ]

#Q. What would be the unit digit in the answer of: $9326^{397} + 1475^{363}$

$6 + 5 = \boxed{1}$

0 1 5 6

$6^1 \rightarrow 6$

$6^2 \rightarrow 6$

$6^3 \rightarrow 6$

4 9

odd / even odd / even

4 6 9 1

2 3 7 8

$5 \rightarrow 5$

$5^2 \rightarrow 5$

$5^3 \rightarrow 5$

$5^4 \rightarrow 5$

1 $4 \rightarrow 4$ R=1

2 $4 \rightarrow 6$ R=0

3 $4 \rightarrow 4$

4 $4 \rightarrow 6$

[MCQ]

#Q. What is the greatest number of 4 digits that when divided by any of the numbers 6, 9, 12, 17 leaves a remainder of 1 ?

A. 9997

B. 9793

C. 9895

D. 9487

9793

612

1224

1836

$612 \times 16 = 9792$

$+1$

$$\begin{array}{r|l} 2 & 6, 9, 12, 17 \\ \hline 3 & 3, 9, 6, 17 \\ \hline & 1, 3, 2, 17 \end{array}$$

$2 \times 3 \times 3 \times 2 \times 17$

$= 612$

Divisibility Rules:

48937 ×

4+8+9+3+7 = 31 ×

- 2: → Unit digit OR last digit is 0 OR multiple 2
- 3: Sum of digits of the given number should be divisible by 3 → 4 → 2
- 4: Last two digits 00 OR Multiple of 4

$8 \Rightarrow 2^{3}$

Last three digits $\underline{000}$ OR

Multiple of 8.

Divisibility Rules:

25 → 5 ②

last two digit of given no. 00 OR Multiple 25

- 5: → last digit of the given number is 0 OR Multiple of 5
- 6: = 2 × 3
- 7:

437154

$\times$ 4893717

489

48 ~ 18 = 30

3 ~ 0 = 3

4892 ~ 2

= 4890

489371 ~ 14

= 489357

48935 ~ 14

= 48921

Twin Prime & Co-Prime Numbers:

5, 7

11, 13

17, 19

3, 5

29, 31

$W_{16} \Rightarrow 1, 2, 4, 8, 16$

$W_{27} \Rightarrow 1, 3, 9, 27$

1. Any two consecutive numbers
2. Any two different Prime numbers

Divisibility Rules:

- 8: → last three digits 000 OR Multiple of 8
- 9: → Sum of digits should be divisible 9.
- 10:

Divisibility Rules:

- 11:
- 12:
- 14:

3×4

2×7

$1 + 9 + 2 = 12$

$\times 1397$

$3 + 7 = 10$

2

Divisibility Rules:

- 15: 3×5
- 16: $\rightarrow 2^{4}$ last 4 digits 0000 OR multiple of 16
- 18: $\rightarrow 2 \times 9$

Divisibility Rules:

- 20: → 4×5
- 40: → 8×5
- 80: → 16×5

[MCQ]

#Q. Which largest number of 5 digits is divisible by 99 ?

$99 = 9 \times 11$

A. 99999

B. 99981

C. 99909

D. 99990

[MCQ]

#Q. Find the least perfect cube divisible by 2, 3, 4 and 6.

Assignment

A 216

B 1728

C 512

D 360

[MCQ]

#Q. How many natural numbers up to 100 are divisible by both 2 and 3?

Assignment

A. 13

B. 14

C. 17

D. 16

[MCQ]

#Q. Find the largest four digit number that is exactly divisible by 88.

Assignment

A 9844

B 9768

C 8894

D 9944

2 mins Summary

Topic ~~Time and Work~~

Number System

THANK - YOU